

Lullaland: A Multisensory Virtual Reality Experience to Reduce Stress

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ABSTRACT

People with health anxiety often experience feelings of nervousness or panic while in high-stress environments. While various techniques such as aromatherapy, and therapeutic video games have been shown to be effective in reducing anxiety, few studies have explored the use of a combination of these techniques in a multisensory approach. In this paper, we propose "Lullaland", a therapeutic virtual reality game that incorporates a wireless diffuser to provide an immersive, multisensory experience designed to help reduce anxiety in medical waiting rooms.

CCS CONCEPTS

• Human-centered computing → Collaborative interaction.

KEYWORDS

virtual reality, aromatherapy, wearable, HCI

ACM Reference Format:

Bo Liu, Wenyu Wang, Yuqing Zhang, Rui Huang, and John Raiti. 2023. Lullaland: A Multisensory Virtual Reality Experience to Reduce Stress. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3585636>

1 INTRODUCTION

People with nosocomophobia, claustrophobia, and agoraphobia often fear or avoid going to places that may trigger panic or feelings of being trapped or embarrassed. For some people, the hospital is associated with things they find stressful and scary, and a negative or traumatic experience can result in additional issues such as trypanophobia, claustrophobia, and latrophobia [27, 39]. Among patients, 15 - 30 % of them have white-coat hypertension, a condition in which a patient's blood pressure readings are higher when

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585636>



Figure 1: We created a portable diffuser that can be used in conjunction with a VR device to enhance the immersive experience and alleviate stress. In this image, A person is using VR device and wearing the diffuser on his sweatshirt, and the background shows a virtual forest scene.

taken at the doctor's office compared to other settings [10]. This indicates that many patients have higher levels of stress and emotional reactions such as anxiety, aggression, and anger when they are in healthcare settings, mainly because they lack control of their environment [26]. These emotions can lead to increased pain sensitivity, delay important medical treatment, and lower the overall efficiency of the healthcare system. Additionally, it can also make patients less willing to continue with their healthcare treatment. Despite these negative effects, hospitals often lack effective methods to distract patients from their anxious emotions and provide a patient-centered, friendly approach to reducing waiting times [1, 41].

Walkthrough the immersive VR experience

Virtual Reality (VR) distraction therapy has been increasingly used to support emotional treatments [8, 17, 24]. Immersive experiences



Figure 2: The Lullaland VR interfaces

make people more accepting of treatment for anxiety and increase treatment effectiveness [5]. However, since the virtual experience is delimited to two sensory inputs (vision and sound), full engagement is delimited. We take inspiration from previous work such as Temperature Illusion [6], Season Traveller [33] and which both incorporate olfactory stimuli to enhance the audio-visual VR technology into a multisensory interactive experience. We also see a graspable olfactory display that can activate scent release by users [31]. However, these designs primarily focus on enhancing the virtual experience, and do not specifically aim to reduce stress in high-anxiety environments. They lack the use of visual and aural stimuli to distract patients from real environments and guide them to a relaxing virtual world [11].

To address the problem of stress and anxiety in hospital waiting rooms, we designed "Lullaland," a multisensory VR game that combines a wireless diffuser with virtual scenes to provide a fully immersive experience to relief stress (Figure 1). The search-and-find game experience increases curiosity about VR scenes, enhancing engagement and relaxation through auditory, visual, and olfactory immersion. Lullaland differs from other passive diffusers as it adapts active aroma designs. It can adjust the type and intensity of aromas based on the virtual environment, making the experience more engaging and effective. Additionally, the use of specific scents can evoke past memories and experiences, which play a key role in anxiety therapy to buffer stress responses [14, 37]. In order to broadly help people in hospital waiting rooms, we designed our product to include:

- (1) A user-friendly interface for people of all ages;
- (2) A portable and lightweight diffuser that can be worn on clothing;
- (3) A controller to help users to navigate the game and reduce movement in hospital waiting rooms.

The gameplay mechanics are designed to be engaging and therapeutic, with the goal for each scene being to find an animal character hiding in a natural environment. On the onboarding page, users will stand in a cabin where they can choose a scene from a list of options (Figure 2(a)). After selecting a scene, they will jump into the scene with a task introduction (Figure 2(b)). The task board describes the target animal's shape and the aroma related to the target animal. As users wander around the scene and look for the animal, the game provides hints through voice or subtitles (Figure 2(c)), and it guides users to find the creature by adjusting the fan speed to control the odor strength. As they approach the animal character, the odor strength automatically increases depending on the distance between the user and the animal character. Once the

user sees the animal character in the environment, they can walk closer to it and click on it to accomplish the goal (Figure 2(d)). The game will then tell users that they have finished the task, and they can continue to walk around for pleasure. Through this process of exploring the scene and finding the animal, users can feel relaxed through a visual and olfactory immersive experience.

2 RELATED WORKS

Traditionally, healthcare waiting rooms offer amenities such as television, soft music, magazines, and more space between seats to help patients reduce stress and make the waiting experience more tolerable [4, 15, 28, 32]. However, playing inappropriate or negative media content in healthcare waiting rooms can make the experience more stressful for patients. Alternative treatments such as meditation and aromatherapy have been found to effectively reduce anxiety in hospitals [3, 25]. As non-invasive, easily applicable, and cost-effective nursing interventions for patients, these methods could be applied in waiting rooms [18]. However, these treatments are typically administered by nurses or psychologists, which may not be commonly available in waiting rooms [2, 16].

VR distraction has become increasingly popular in recent years, particularly in the management of acute pain [11]. VR videos of natural scenes with a relaxing soundtrack can make patients feel like they are in a calm and peaceful place. Research has shown that natural environments such as forests or coasts can be more effective in reducing stress and anxiety than urban environments [13, 19]. The sense of presence and immersion that VR provides makes it more effective in reducing stress than other technological interventions [7]. However, the VR immersive experience can be easily disrupted by the hospital smell of disinfectant [23]. The hospital smell can become associated with negative emotions and memories of painful treatment and illness [20]. To effectively relieve patients' emotions in such a setting, VR devices should provide a multisensory experience. Dozio et al. found that multisensory stimuli play a key role in driving participants' attention into the virtual world compared to single sensory stimuli [9]. Multisensory experiences can provide a more realistic user experience and increase immersion and engagement [29].

Previous studies, such as Temperature Illusion and Season Traveller, have demonstrated the potential of incorporating olfactory stimuli in VR scenes by using scents to activate the olfactory bulbs and trigeminal nerve, resulting in more realistic and engaging virtual experiences. For instance, Temperature Illusion uses specific scents to simulate warm and hot temperatures [6]. However, these

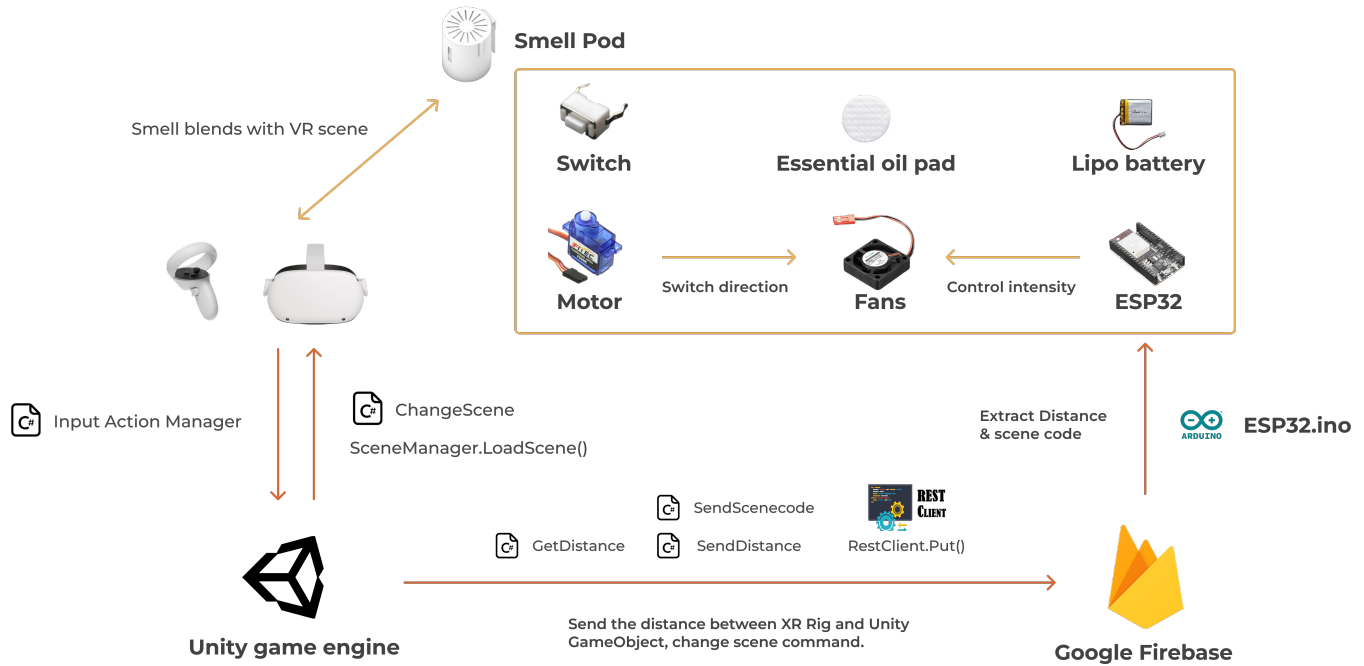


Figure 3: Hardware and Software Architecture

designs have limitations, such as a lack of engaging gameplay mechanics and natural, stress-relieving environments, which could further enhance the therapeutic benefits of VR experiences.

3 SYSTEM OVERVIEW

We developed an interactive game experience using the Oculus Quest 2¹ and Unity game engine². Users interact through touch controllers to move or turn in the game scene and the aroma diffuser adjusts odor strength automatically to guide users in the game.

Our minimum viable product [35] version chose forest and beach as two calming locations, as most people feel relaxed and pleasant with natural places [12, 21, 38]. Based on the scenes, we selected white lavender and citronella scents for the forest scene and coconut scent for the beach scene. These scents are chosen based on the scenes and have effects to help people relief [22, 30, 36]. We also present our system architecture in Figure 3.

3.1 Hardware Implementation

Figure 4 shows the design structure of the diffuser. We 3D printed a cylinder for rotating purposes, which measures 28 mm in radius and 68 mm in height. The core of our device is a combination of three fragrance pods that can be switched out and the fan speed can be adjusted. Our wireless diffuser consists of two fans, a motor connected with a turn panel, a battery, and a control board (ESP32). The ESP32 connects to the cloud data and controls the board, motor, and fans to rotate, turn on or off. The fragrance oil pod container has four isolated sections to hold fragrance oil pods, which can easily be removed from the container. In the future, users will be

able to choose the scents and the amount of fragrance oil on the pod based on their preference. Three sections were designed with air outlets to spread the scents, and one section is designed to be fully blocked. The turn panel vent stays at the blocked section when the users are not playing the game. When users enter a scene, the turn panel vent will switch to the corresponding pod container and fans begin to work. On the container, we designed airflow vents and a stabilizing clip for secure positioning. Additionally, we chose two different sizes of fans for four-speed adjustments, which will adjust the odor strength. This design allows for a customizable and user-friendly diffuser experience.

3.2 Software Implementation

First, we built two different scenes in the Unity game engine: the forest³ and the beach⁴. We used the XR interaction toolkit package⁵ to develop the interaction. A locomotion system was added to the terrain so that when the game begins in Oculus, users can navigate and explore the virtual world using the left and right controllers to interact. Furthermore, as the user approaches the animal character, we calculated the distance between the user (XR Rig) and the animal character (Unity Gameobject) in the scene.

To control the aroma diffuser, we used an open-source package called Restclient⁶ in Unity to send data to Google Firebase. As users navigated the scene, we sent the distance data and the current scene code to the cloud. Then, in the aroma diffuser, the ESP32 extracts

¹<https://www.oculus.com/>

²<https://unity.com/>

³<https://assetstore.unity.com/linkmaker/embed/package/152682/widget-wide?aid=101119e3f&pubref=website>

⁴<https://assetstore.unity.com/packages/3d/environments/fantasy/hand-painted-island-pack-36959>

⁵<https://github.com/Unity-Technologies/XR-Interaction-Toolkit-Examples>

⁶<https://github.com/rest-client/rest-client>

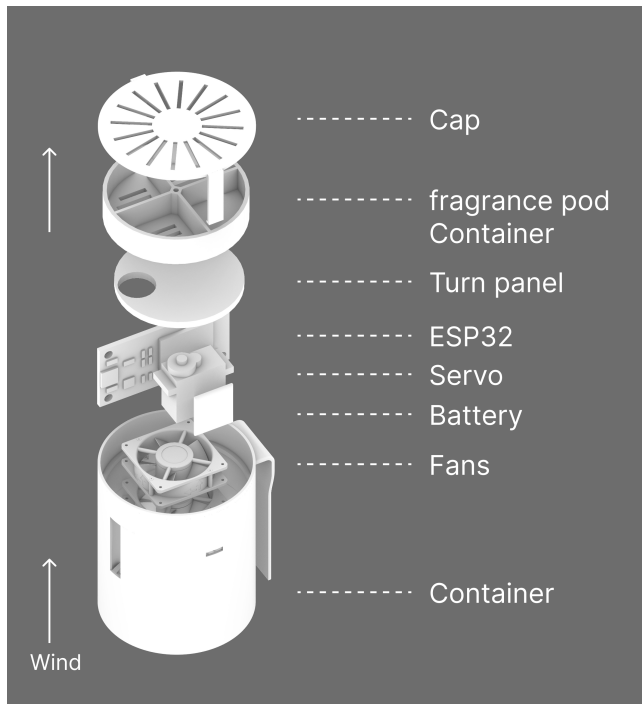


Figure 4: Diffuser Design Structure

the distance data to control the fans and adjust the odor's intensity. Moreover, the scene code was used to control the spin angle of a motor to adjust the smell it diffuses. Therefore, the smell blends with the scene the user sees in Oculus. This way, the aroma diffuser is synced with the scene in real-time, providing a more immersive and realistic experience for the user.

4 PILOT STUDY

We conducted a pilot study within the research team using Lullaland to gather early-stage feedback on the VR interaction and anxiety levels of participants before and after using our product. This stage of usability test will help us to find potential bugs and validate our hypothesis. Our main hypothesis was that people with anxious mood in high-stress environments would experience a decrease in anxiety after playing our game. We believed that Lullaland's multisensory natural scene VR experience would allow users to fully engage in the game and find relief from the stress of their environment. Our second hypothesis was that by adjusting the strength of the smells, users would enjoy the process of finding animals and get hints from it. The study aimed to test the effectiveness of our VR game in reducing anxiety levels and enhancing the user's overall experience through the use of olfactory stimuli.

Task and Procedure

At the beginning of each pilot study, we provided context to the participants by having them imagine that they have just checked-in at a clinic and are currently in a waiting room for their doctor's appointment. To establish a baseline measurement, participants were then asked to complete an anxiety assessment. After completing the

assessment, participants were given the option to choose between two VR scenes, a forest and a beach scene, to explore. Each scene has a task introduction that guides participants to find a hidden animal character by following hints and using their sense of smell. Participants were also instructed that they can exit the game at any time in case they are called by their doctor for their appointment. Upon completion of the task or exiting the game, participants were asked to complete another round of anxiety assessment to measure the change in their anxiety levels before and after the multisensory VR experience.

We had 4 participants (2 female, 2 male, 24 ± 3 years old) for our pilot study. All participants had VR experience but no participants had aromatherapy before. In order to accurately collect responses and gain feedback, no participants previously played the game. No participants report allergies to fragrance pods' ingredients.

Result

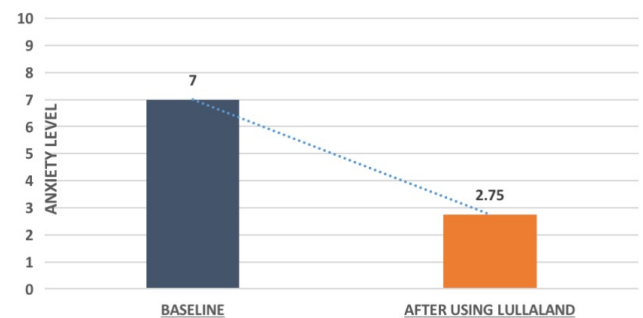


Figure 5: Anxiety level changes between before and after using Lullaland

Our pilot study of Lullaland revealed a significant decrease in anxiety levels among participants who used our multisensory VR device. To measure anxiety levels, we asked participants to rate their level on a scale of 1 to 10, with 1 indicating minimal anxiety and 10 indicating extreme anxiety. After we asked participants to imagine they are sitting in a hospital waiting room, the average anxiety level for the four participants was 7 before using the device. After participants used our device to complete tasks and experienced virtual scenes, their anxiety level decreased to 2.75, as shown in (Fig 5). Participants on average failed 0.75 out of 4 tasks, indicating that while they found the experience relaxing, they could not complete all the tasks. Two participants accidentally quit the game due to their unfamiliarity with the Oculus device, and one participant did not finish reading the onboarding instructions. The onboarding instruction board was set to disappear after 10 seconds, which may have contributed to these issues. Despite these challenges, all four participants expressed satisfaction with their overall experience.

5 DISCUSSION AND LIMITATION

Our pilot study of Lullaland demonstrated its potential to help people engage in the game and reduce anxiety in various settings,

such as hospital waiting rooms. Our results showed that Lullaland (i) relieved users' anxiety levels after playing the game and (ii) guided users to complete the game task through changes in smell strength and hints. However, during our pilot testing, we received several suggestions from participants on improving our prototype. Some participants found it difficult to understand the connection between the animal and the scent at the beginning of the game. For example, we used lavender and citronella scents for the forest scene to guide participants to finish the task; these scents were chosen to fit the scene. But it is not directly relevant to the scene's task animal (fox). Additionally, participants enjoyed the search process but wished to have a more complicated design to engage more in the game. For example, a person suggested having multiple tasks in one scene or adjusting the task difficulty as an option.

A major limitation of our evaluation is the anxiety assessment and anxiety measurement. Although previous studies have validated that VR scenes and scents can relieve anxiety, we lack a no-odor baseline condition to determine how the smell impacts relaxation. With this baseline, we can better measure participants' anxiety levels after using Lullaland. We conducted this study in a school conference room, and participants who have emotional changes in a high-anxiety environment may have different anxiety assessment results compared to the results we obtained in our pilot study. To collect feedback accurately, we also need to expand our usability study to a large group of people outside the institution.

6 CONCLUSION AND FUTURE WORKS

In this paper, we proposed Lullaland, a multisensory VR game, and custom diffuser system, as a solution to reduce stress in high-anxiety environments. We believe this system can effectively help people engage in the game and relieve anxiety in various settings, such as hospital waiting rooms. With the fast-growing technology and the increasing need for better care experiences, there are many possibilities and demands for the discovery of distraction-based VR interactions in healthcare, airports, and among people with mental disorders [34, 40].

In future works, we plan to continue iterating based on usability results, such as adding more scenarios and game levels to provide a better user experience and selecting fragrance scents that match scenes and are tied logically to task animals. We will also conduct usability tests to further evaluate the effectiveness of our Lullaland system from various angles: measuring the effect with no-odor control, testing in healthcare environments to see the changes in stress levels of patients in actual cases, and determining the system's feasibility for deployment in most hospitals, increasing the sample size of participants to validate the effectiveness of Lullaland in reducing anxiety levels, and collecting users' biosignal data using wearable sensors and analyzing users' stress levels to validate the effectiveness of our system further.

7 ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to Professor Sarah Zuberec and teacher assistant Haonan Peng for their invaluable advice and support throughout the courses. Their guidance and feedback have been instrumental in shaping our research and improving the quality of our paper.

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